

System Installation (Deployment) Manual – LIDA Composting System

1. Introduction

This document provides the technical setup and deployment procedures for the LIDA Composting System. It outlines the installation of both the 3D model and the mobile application, configuration of the sensor simulation environment, and linkage between the two platforms for testing and demonstration.

2. System Components

The system consists of the following components:

- 1. 3D Model Environment (SolidWorks/SketchUp/Unity):
 - Includes composting chambers, pipeline system, water tank, and internal paddles.
- 2. Sensor Simulation Layer:
 - Gas, temperature, and moisture sensors for live data representation.
- 3. Mobile Application (Flutter):
 - Displays real-time composting data, animations, and feedback.
- 4. Communication Layer:
 - Optional linkage through MQTT or local data exchange for visual updates.
- 5. Monitor Display Integration:
 - Virtual display inside the 3D model to simulate on-unit feedback.

3. Hardware and Software Requirements

Hardware

Component	Minimum Requirement
Processor	Intel i7 or equivalent
RAM	8 GB or higher
Storage	2 GB free space
Display	1080p or higher resolution
Mobile Device	Android 10.0+ or iOS 14.0+

Software

Software	Version / Notes
SolidWorks / SketchUp	2024 or newer
Unity / Blender (optional)	For animation and simulation
Flutter SDK	Latest stable version
Visual Studio Code / Android Studio	For mobile app deployment
Python / Node.js (optional)	For data simulation scripts

4. Installation Steps

A. Setting up the 3D Model Environment

- Download and extract the LIDA_Composting_Model.zip package.
- Open the main assembly file (LIDA_Model.SLDASM or .SKP) in SolidWorks or SketchUp.
- Verify that all references and components are properly linked (chambers, paddles, pipes, tank).
- Enable the animation manager to activate motion for paddles, airflow, and condensation.
- Export the file as .GLB or .FBX for mobile app integration testing.

B. Installing the Mobile Application

- Open Android Studio or VS Code.
- Clone or extract the Flutter project folder (lida_app).
- Run the following commands:

```
flutter pub get  
flutter run
```

- Connect a compatible device or use an Android emulator to launch the app.
- Ensure the app displays simulated data fields for temperature, gas, and moisture.

C. Linking the 3D Model with the Mobile Application

- Copy the exported .GLB model into the Flutter project's assets/models/ folder.
- Update the app configuration file.
- Verify the 3D rendering inside the mobile interface.
- Confirm that the model rotates, scales, and displays interactive elements smoothly.
- Test real-time data display using mock sensor values.

5. Data and Sensor Simulation Setup

1. Enable mock sensor data via integrated scripts or local JSON simulation files.
2. Ensure the app reads these values and refreshes the displayed data periodically.
3. Use placeholder variables for:
 - Gas: CO2_level
 - Temperature: tempC
 - Moisture: humidity
4. Test sensor accuracy by triggering mock variations and observing visual changes in the composting animation.

6. Testing the Deployment

- Visual Test: Confirm that animation and color-coded chambers (red—active, green—curing, blue—finished) are visible.
- Data Test: Ensure all sensor readings are updating dynamically in the mobile app.
- Performance Test: Exported GLB should run smoothly on mid-range mobile devices without lag.
- Integration Test: The monitor display inside the 3D model should correspond with app data.

7. Troubleshooting Guide

Issue	Possible Cause	Solution
App cannot load 3D model	Asset path incorrect	Verify model path in pubspec.yaml
Sensor data not updating	Simulation script error	Restart local mock data server
Animation not playing	Disabled animation in viewer	Enable animation playback in model settings
Drainage logic missing	Layer visibility disabled	Turn on water tank and pipe layers

8. Maintenance and Updates

- Update the Flutter app dependencies regularly with flutter pub upgrade.
- Keep SolidWorks/SketchUp libraries updated for compatibility.
- Maintain clear documentation for new hardware integration (e.g., motors or live sensors).
- Ensure new visual elements (e.g., monitor display) are properly layered and labeled.

9. Ethical and Cybersecurity Considerations

- Ensure compliance with Australian Privacy Principles (APPs) for data protection.
- Avoid collection or transmission of personal or identifiable data.
- Use secure encryption protocols (HTTPS/TLS) if data is transmitted between systems.
- Label simulated data to avoid confusion with real-world readings.

10. Conclusion

This deployment manual provides step-by-step guidance for setting up, linking, and testing the LIDA Composting System. Following these procedures ensures that both the 3D model and mobile application function in synchrony, demonstrating airflow, condensation, and composting logic accurately and securely.